Repo link : https://github.com/AD12334/PatchTrack

| When | Stage | Minimum record |
|---|---|---|
| Week 5 – M1 | Study + RQs + preregistration | Record/freeze RQs, bounded target, sample IDs, method, metric/decision rule, expected result and what would count as non-reproduction. Tag: M1_PLAN. |

| | |
|---|---|
| Research Question 1 | To what extent can the PatchTrack classification results (the distribution of Patch Applied / Patch Not Applied / No Patch Exists PRs, and the median integration rate) be reproduced from the authors' replication package, and how robust are they under an independent audit? |
| Source anchor | 4.2 - The paper does a qualitative analysis of the sampled cases; they look at how much of the code is integrated into the codebase. The patches were classified into six different classes each describing how much of the AI generated code had to be changed |
| Target result | Reproduce results achieved in the research that AI-assisted code has a median 25% adoption rate. |
| Out of scope | Examining PR's that don't explicitly disclose AI usage. |

| | |
|---|---|
| Unit of analysis | Individual GitHub pull requests containing self-admitted ChatGPT usage. |
| Source/version | The dataset was released with the source study containing pull requests with self-admitted ChatGPT usage. |
| Size/range | 80 pull requests from the released dataset. |
| Selection rule | the first 80 eligible items in stable order (chronological order) |

| | |
|---|---|
| Input (sample) | 80 selected pull requests from the study's released dataset containing self-admitted ChatGPT usage. |
| Procedure | • Load PatchTrack tool<br>• Use patch track tool to examine AI-code adoption rate per PR<br>• Compare average adoption rate with paper findings |

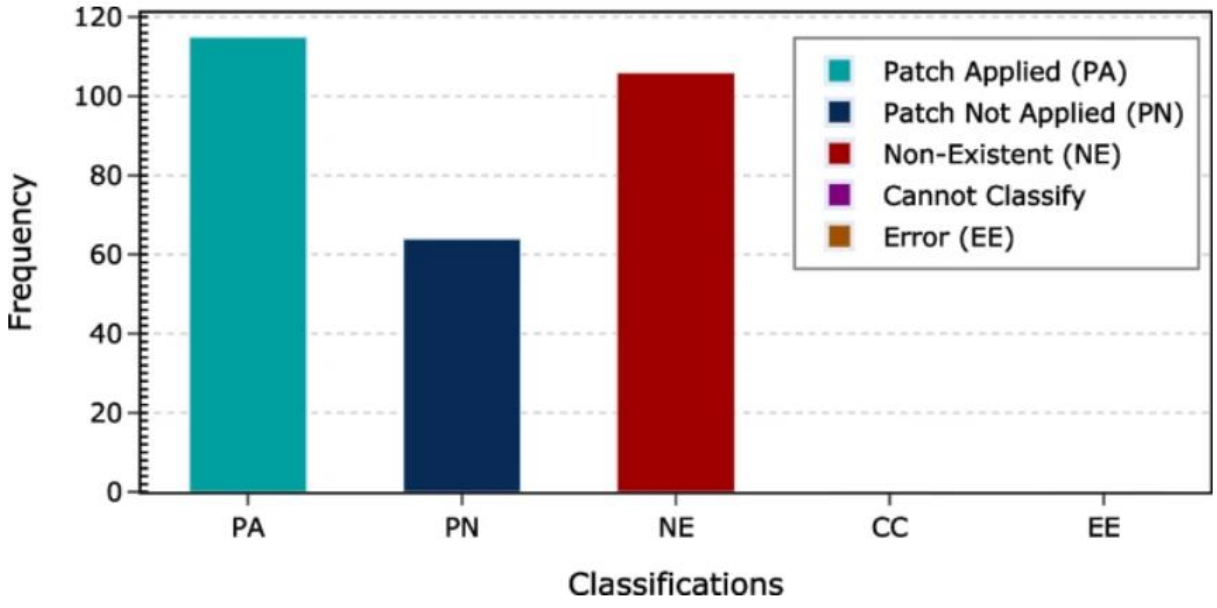
| | |
|---|---|
| Output | A comparison between paper findings and independent findings. |

| | |
|---|---|
| REPRODUCED | The observed acceptance rate matches the source study's reported acceptance rate according to the predefined comparison criteria. |
| NOT REPRODUCED | The observed acceptance rate does not match the reported acceptance rate according to the predefined comparison criteria |
| INCONCLUSIVE/BLOCKED | BLOCKED |

| | | |
|---|---|---|
| Week 7 – M2 | Bounded reproduction + audit | Complete the bounded reproduction or document a defensible infrastructure limitation using the contingency route. Record raw/result evidence and key deviation/threat. Tag: M2_REPRO and PRE_EVENT_A |

Q: To what extent can the PatchTrack classification results (the distribution of Patch Applied / Patch Not Applied / No Patch Exists PRs, and the median integration rate) be reproduced from the authors' replication package, and how robust are they under an independent audit?

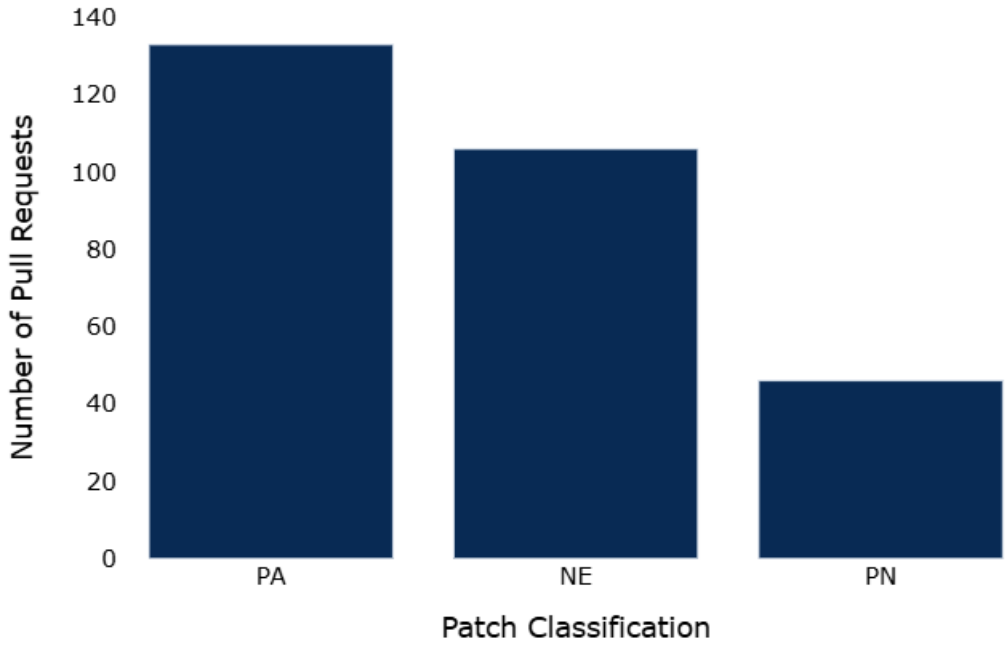
The report investigates the usage of ChatGPT -assisted pull requests across three integration categories: Patch Applied (PA ), Patch Not Applied (PN ), and No Patch Exists (NE ). The paper investigated 285 pull requests and featured the following distribution of results.



- **Patch Applied (**PA **)**: 116 cases (40.7%)
- **Patch Not Applied (**PN **)**: 63 cases (22.1%)
- **No Patch Exists (**NE **)**: 106 cases (37.2%)

Under our bounded investigation, we ran Patchtrack on the first 80 (28% sample) pull requests featured in the paper's dataset. In our investigation, we achieved the following results. “PA 42 (52.5%) NOT EXISTING 26 (32.5%) PN 12 (15%)”.

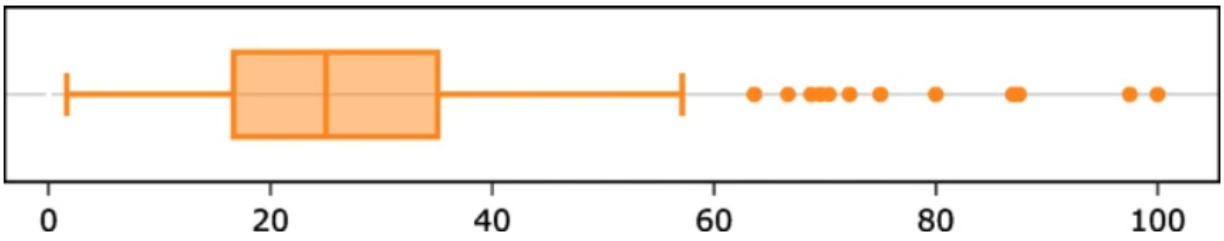
```
Patch Classification
PA    42
NE    26
PN    12
Name: count, dtype: int64
```



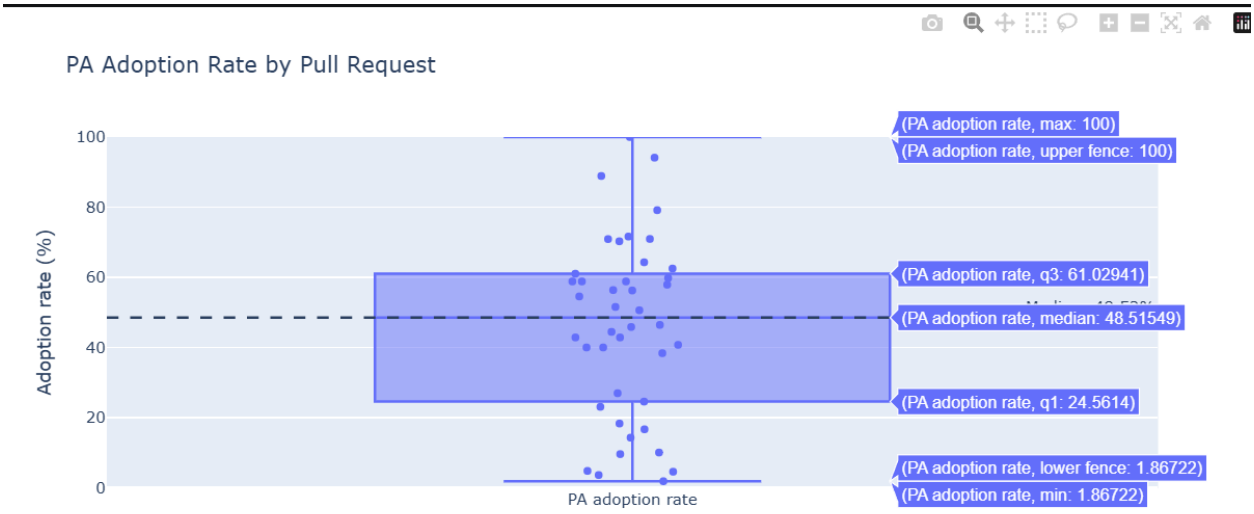

Furthermore, the paper uses “Jaccard’s containment ratio 1 to quantify the overlap between tokens in the ChatGPT-suggested patch () and the final integrated code ()”.

$$Jacc(t_x, t_y) = \frac{|t_x \cap t_y|}{|t_x|} \times 100$$

Where Tx and Ty are “token streams from ChatGPT and GitHub patches, respectively”. The paper states a median adoption rate of 25 %, along with a third quartile at 38%.




When comparing to our bounded sample we observed a median adoption rate of 48.52% along with a third quartile at 61%




## Conclusions

When comparing our bounded results to the results achieved in the paper, we observe the following.

1) **Similar patch classification results:**

We got an 11.8% increase in PA, a 4.7% decrease in NE and a 7.1% decrease in PN. For testing, we used a sample size of 80 out of the original 285. We chose a sample of this size because it gave was large enough to give us information on the classification distribution and PA adoption rate whilst staying small enough for us to run PatchTrack

2) **Median patch adoption rate varies significantly**

When we ran PatchTrack we got a median adoption rate of 48.5% with a third quartile of 61%. This is nearly double the papers median adoption rate of 25% and a 23% increase for the third quartile. We feel that this might have occurred due to a sample bias. As stated earlier, we only had a sample of 80 out of 285. That's roughly 28% of the original sample. It may have also occurred due to a different population. If the papers 25% consisted of repos where the code wasn't adopted then the median would be pulled down. From this we can conclude that our median adoption rate was not reproduced.